Towards Robust Video Anomaly Detection using a Multi-Task Learning Approach

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Abstract

In today's world, ensuring safety and privacy is essential. Detecting anomalies in video is vital for providing security in various domains, such as public surveillance, transportation, and healthcare, as these unexpected events can indicate potential threats or emergencies. Video anomaly detection focuses on identifying events that diverge from typical or expected patterns of behavior. Traditional methods often rely on labeled datasets, which can be difficult to obtain and may not effectively integrate both spatial and temporal dynamics, leading to limitations in identifying subtle anomalies. To address these issues, we propose a novel approach combining frame prediction with self-supervised and contrastive-based multi-task learning. Our method leverages three complementary tasks—future frame prediction, classification, and similarity measurement to develop robust representations of normal behavior while explicitly capturing spatio-temporal dynamics. This unified framework enhances robustness and generalization to unseen anomalies, particularly in scenarios with limited labeled data. By effectively combining these methodologies to improve anomaly detection performance our approach sets itself apart from conventional supervised and unsupervised models

1. Introduction

Anomaly detection in videos refers to the identification of events or behaviors that deviate from what is considered normal or expected [5]. This task is critical in a wide range of applications, particularly in video surveillance, where detecting unusual activities, such as accidents, criminal actions, or unsafe behaviors, is of utmost importance. Other use cases include industrial monitoring, healthcare (for patient monitoring), autonomous driving, and smart city management.

Despite its importance, anomaly detection remains a highly challenging problem. The primary reason for this is the nature of abnormal events, which are often unbounded and diverse.[18] Unlike normal events, which typically follow regular patterns and can be easily characterized, abnormal events are rare, unpredictable, and can vary widely depending on the context. For instance, an anomaly in a pedestrian crossing may involve a person running, whereas an anomaly in a factory setting may involve a machine malfunctioning. This variability makes it extremely difficult to model anomalies explicitly.

As a result, unsupervised or semi-supervised approaches are often employed for anomaly detection in videos. In these approaches, models are typically trained on normal data only. The underlying assumption is that the model will learn to recognize patterns of normal behavior and then identify deviations from these patterns as anomalies. Various techniques, such as feature reconstruction, prediction-based methods, and more recently, deep learning methods like Autoencoders, Generative Adversarial Networks (GANs), and contrastive learning, have been developed to address this problem. These methods aim to capture the inherent characteristics of normal events, either by predicting future frames, reconstructing the input data, or distinguishing between similar and dissimilar events in the learned feature space.[6]

Nonetheless, anomaly detection in videos still faces significant challenges, such as the spatiotemporal complexity of video data, the lack of labeled anomalies, and the need for models that can generalize to new types of anomalies. These challenges make it a compelling area of research, with ongoing efforts to develop more robust and scalable solutions.

In this work, we want to present a novel methodology for video anomaly detection that utilizes future frame prediction in conjunction with self-supervised and contrastive learning methods allowing the model to learn without the need for explicit labels. Our approach addresses the limitations of traditional methods that often rely on labeled datasets, which can be scarce and challenging to obtain.
Our approach involves three key components:
1. Future Frame Prediction: By predicting subsequent frames based on the current frame, our model captures the temporal dynamics essential for understanding video content. This predictive capability allows us to establish a baseline of expected behavior, against which anomalies can be detected.
2. Self-Supervised Learning: Self-Supervised learning method leverages unlabeled data by designing pretext tasks, enabling the model to learn meaningful feature representations without requiring extensive labeled datasets. This is particularly advantageous for anomaly detection, where anomalies are rare and manually labeling every possible scenario is impractical.
3. Contrastive Learning: Contrastive learning used to strengthen the model's ability to differentiate between normal and abnormal sequences. By comparing the predicted future frames with the actual frames, the model learns to identify significant discrepancies that indicate anomalies. This process not only enhances representation learning but also improves the model's robustness and generalization to unseen anomalies.

2. Related Work

2.1. Frame Prediction

In the case of video anomaly detection, the main problem is the event will occur rarely which is unpredictable for human to predict. Anomaly detection is scene-dependent which makes it to predict as it changes based on context at a particular time. For Example, driving the car on a road is common but if driving car happens on the pedestrain way makes it abnormal as shown in Figure 1. In previous works, researchers also worked on this problem through unsupervised learning but they lack adequate and flexible generalization of normal events.[9]
In the studies about this topic mentioned in [19] authors have mentioned frameworks like Hand-crafted Features Based Methods [10][20]: i) extracting features either hand-crafted or learned on training set; ii) learning a model to characterize the distribution of normal scenarios or encode regular patterns; iii) identifying the isolated clusters or outliers as anomalies. The fundamental underlying assumption of these methods is that any regular pattern can be linearly represented as a linear combination of basis of a dictionary which encodes normal patterns on training set. Therefore, a pattern is considered as an anomaly if its reconstruction error is high and vice versa. However since the dictionary is not trained with abnormal events and it is usually incomplete, we cannot guarantee the expectation. Deep Learning-based Methods [7][14][21]: they usually learn a deep neural network with an Auto-Encoder way and they enforce it to reconstruct normal events with small re-construction errors. However, all these anomaly detections are based on the reconstruction of regular training data, even though all these methods assume that abnormal events would correspond to larger reconstruction errors, due to the good capacity and generalization of deep neural networks, this assumption does not necessarily hold. Therefore, reconstruction errors of normal and abnormal events will be similar, resulting in less discrimination. Since anomaly detection is the identification of events that do not conform the expectation, it is more natural to predict future video frames based on previous video frames, and compare the prediction with its ground truth for anomaly detection and the proposed model here is slightly modified U-Net architecture for generating predicted future frames, to make the prediction close to its ground truth i.e. predicted frame(t+1) and ground truth frame(t+1), intensity and gradient difference are used, however only appearance constraints cannot guarantee to characterize the motion information well. Besides spatial information, temporal information is also an important feature of videos.

In [12], authors introduced a novel model for anomaly detection which is inspired by [3]However, [12] has limitations. The model is trained on specific datasets, generalization of detecting anomalies is not fully explored.

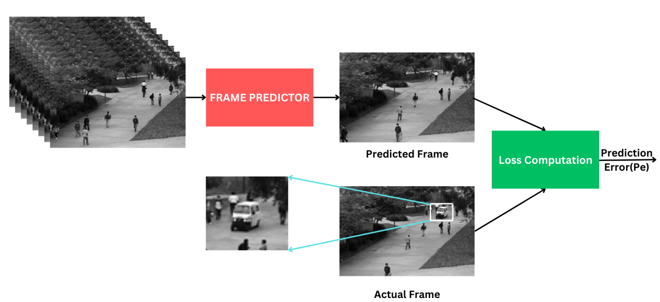

Figure 1. Frame prediction task for anomaly detection.

As deep learning techniques emerged, approaches like autoencoders[17][8] addressed some limitations by learning complex patterns from normal data but were hampered by their static architectures, limiting adaptability to varied anomalies. Subsequent works[11], such as those utilizing generative adversarial networks (GANs), sought to create synthetic anomalies, yet they still faced challenges in accurately modeling real-world scenarios. But They work well only for short-term anomalous movements, but they lack an understanding of normality. But the data-driven techniques like deep learning make it hard to reconstruct the normal routine events that occur rarely.

In [4], the author proposed a model in which they proposed a framework to deal with the above problem. They created a model based on the context recovery method and knowledge retrieval method.The context recovery method modeled the normal motion patterns by predicting the future frame based on the input snippet. The knowledge retrieval method encodes the knowledge about normality and stores in a knowledge base. Finally, they fused two anomaly probabilities and obtained the final anomaly score on testing data. Traditional approaches use a single static network architecture to detect anomalies, making it difficult to capture various complex patterns. To address this, the authors in [24] propose the DSS-Net architecture, which leverages a dynamic self-supervised approach to effectively capture diverse spatial and temporal features, thus enhancing the capability of video anomaly detection in critical applications such as security surveillance and traffic monitoring. But, the pseudo-anomalous data generated during training may not accurately capture the complexities of real-world anomalies, which could affect the model's generalization.

2.2. Self-Supervised Learning

Self-Supervised learning(SSL) particularly advantageous in scenarios where labeled anomalous events are scarce or unavailable. Recent research has introduced various SSL techniques tailored for VAD, each leveraging unique proxy tasks to capture the underlying structure of normal video patterns. In [15] authors provided an extensive review of self-supervised anomaly detection methods, discussing their strengths, limitations, and potential future directions.In [13] authors proposed a multi-task learning approach that integrates multiple self-supervised tasks, such as motion irregularity detection and knowledge distillation, to enhance anomaly detection in videos.Building on the framework,[2]introduced SSMTL++, which incorporates vision transformer modules and additional self-supervised tasks, achieving state-of-the-art performance across several datasets.

2.3. Contrastive Learning

Contrastive Learning creates a discriminative embedding space where similar samples are pulled closer together and dissimilar ones are pushed apart. This method is particularly effective for learning robust representations without labeled data, making it ideal for tasks like anomaly detection, where it helps capture subtle differences by focusing on the intrinsic features of the data. The learned embeddings are highly transferable to downstream tasks, enhancing performance and generalization.

In [25], author proposes a two-stage framework called ICDC for self-supervised video learning. It first enhances Instance-Wise Contrastive Learning (Instance-CL) with a novel consistency-preserving sampling strategy to generate more motion-sensitive video representations. Then, it applies Deep Clustering (DC) to refine these representations, improving intra-class compactness and inter-class diversity. This approach is applied to tasks like action recognition and video retrieval, demonstrating improved performance over state-of-the-art methods without using labeled data.

In [16], author introduces TAC-Net, a temporal-aware contrastive network designed to address challenges in video anomaly detection (VAD). TAC-Net uses deep contrastive learning and multiple self-supervised tasks to improve anomaly detection in surveillance videos. Instead of relying on a single pretext task like frame prediction or reconstruction, it jointly learns multiple tasks, including temporal transformations, to capture high-level semantic and temporal features of normal patterns. This approach reduces issues like the shortcut problem, where models focus on low-level features, and the nonalignment between pretext tasks and anomaly detection. But for long sequence data it may not perform as it is mentioned in the paper.

3. Methodology

To address these challenges, a multi-task learning approach was developed for anomaly detection.Frame Prediction, Self Supervised Learning and Contrastive Learning tasks were considered as in context of anomaly detection these approaches were better than labelled-based methods in real-time applications. The detailed architecture is shown in Figure 2.

3.1. Data Augmentation

Given a frame $I_i = \mathbb{R}^{H \times W \times 3}$.We considered set of t contiguous frames as a snippet S_i and have adopted four temporal transformations T= T1, T2, T3, T4 namely temporal reverse, jittered rotation, speed and shuffle to each snippet. This will generate 4 sets of augmented snippets $S_i^{T1}, S_i^{T2}, S_i^{T3}, S_i^{T4}$ for each given snippet S_i.

Temporal Reverse: This will take the snippet and reverse the order of frames in the snippet. It is specifically used to discriminate whether the temporal order of a video snippet is forward or backward.

Jittered Rotation: Four types of jittered rotations, i.e.,$0^o, 90^o, 180^o$, and270^o for every frame on the raw snippet were applied. Besides, a random noise $\in [3^o, 3^o]$is

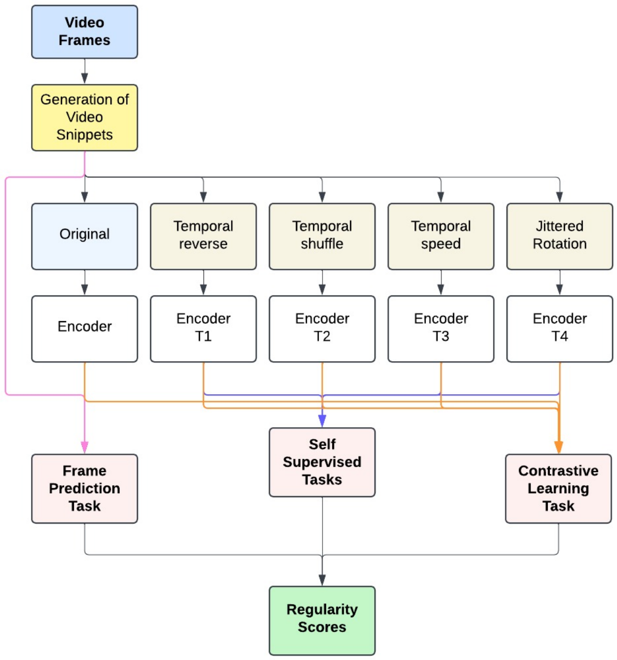

Figure 2. Proposed Architecture

added to the original rotation.
Shuffle: For the Given a video snippet, we randomly shuffle its temporal orders of frames.
Speed: Various sampling rates(range(0.5,2)) on a video snippet to obtain snippets with different playing speed.

3.2. 3D Encoder

Both the raw snippet S_i and its augmented variants $S_i^{T1}, S_i^{T2}, ...S_i^{T4}$ are fed into a 3D-ResNet encoder to extract snippet-level spatiotemporal features $f(S_i)$ and $f(S_i^{T1}), f(S_i^{T2}), ...f(S_i^{T4})$ before passing through encoder the input videos are preprocessed to the resolution of 256 × 256 and the intensity of [-1, 1] including the augmented variants for all datasets. The encoder utilizes 3D convolutional kernels TXS^2, C, where T, S, and C represent the temporal, spatial, and channel dimensions, respectively. The encoder's architecture is detailed in Figure 4. Residual blocks progressively reduce temporal and spatial dimensions using strides r x s^2 while increasing channel dimensions to capture rich spatiotemporal patterns. The output dimension we incur after the encoder task is $R^{N,F,1,32,32}$ where N is number of snippets and F is number of features. The encoder outputs high-level spatiotemporal features, which are then processed to Task Head and projection head. The Network architecture details were explained in Figure 3

Blocks	Architecture	Output $T \times S^2$
conv	$7 \times 7^2, 64$ stride:1,1^2 +ReLu+BN	8×256^2
res1	$\begin{bmatrix} 3 \times 3^2, 128 & \text{stride}: 2, 2^2 \\ 3 \times 3^2, 128 & \text{stride}: 1, 1^2 \end{bmatrix} \times 1$ $\begin{bmatrix} 3 \times 3^2, 128 & \text{stride}: 1, 1^2 \\ 3 \times 3^2, 128 & \text{stride}: 1, 1^2 \end{bmatrix} \times 2$	4×128^2
res2	$\begin{bmatrix} 3 \times 3^2, 256 & \text{stride}: 2, 2^2 \\ 3 \times 3^2, 256 & \text{stride}: 1, 1^2 \end{bmatrix} \times 1$ $\begin{bmatrix} 3 \times 3^2, 256 & \text{stride}: 1, 1^2 \\ 3 \times 3^2, 256 & \text{stride}: 1, 1^2 \end{bmatrix} \times 3$	2×64^2
res3	$\begin{bmatrix} 3 \times 3^2, 512 & \text{stride}: 2, 2^2 \\ 3 \times 3^2, 512 & \text{stride}: 1, 1^2 \end{bmatrix} \times 1$ $\begin{bmatrix} 3 \times 3^2, 512 & \text{stride}: 1, 1^2 \\ 3 \times 3^2, 512 & \text{stride}: 1, 1^2 \end{bmatrix} \times 5$	1×32^2

Figure 3. Encoder Architecture

3.3. Frame Prediction

We adopt an improvised approach to the frame prediction task, leveraging raw snippet-level features ($f(S_i)$) for anomaly detection. Unlike the conventional decoder-based reconstruction of all frames within a snippet, our method focuses on predictive modeling by utilizing the initial frames of a snippet to predict the subsequent frame. This predictive approach not only simplifies the model's output but also provides a more targeted framework for anomaly detection. Our decoder architecture adheres to a structure composed of three upsampling layers and three residual blocks with 2D convolutions. Each upsampling operation employs nearest-neighbor interpolation to incrementally restore spatial resolution. The final convolutional layer is used to reconstruct the RGB frames of the video snippet. In this model, we considered there are 8 frames per snippet. Hence from the reconstructed frames of the video, we take first 7 frames and predict the 8^{th} frame. Let $h_{T_{pre}}$ be the frame prediction task head. To further refine the prediction quality, we incorporate intensity and gradient constraints (L_{int} and L_{grd}) to the loss function. These constraints enhance the model's ability to preserve fine-grained spatial and temporal details, ensuring that both high-level semantics and low-level features are accurately represented in the predictions.

$$
\begin{aligned}
L_T^{pre} &= L_{int} + L_{grd} \\
L_{int}(S_i, Y^{T_{pre}}) &= \frac{1}{WH} \sum_{j=1}^{W} \sum_{k=1}^{H} \| Y_{jk}^{T_{pre}} - \bar{Y}_{jk}^{T_{pre}} \|_2^2 \\
L_{grd}(S_i, Y^{T_{pre}}) &= \frac{1}{WH} \sum_{j=1}^{W} \sum_{k=1}^{H} \Big(\| \nabla_x Y_{jk}^{T_{pre}} - \nabla_x \bar{Y}_{jk}^{T_{pre}} \|_1 \\
&\quad + \| \nabla_y Y_{jk}^{T_{pre}} - \nabla_y \bar{Y}_{jk}^{T_{pre}} \|_1 \Big)
\end{aligned} \tag{1}
$$

Where $\bar{Y}^{T_{pre}} = h_{Tpre}(f(S_i))$ indicates the predicted frame, $Y^{T_{pre}}$ represents the actual 8^{th} frame in the generated snippet, i, j represents spatial index, W, H indicates width and height of the frame. ∇_x and ∇_y denote the gradient along x-axis and y-axis.

In the overall, the decoder process the input $R^{N,D,1,H,W}$, where N represents the number of snippets, D represents the embedding dimension, and (H, W) represents the spatial resolution of the encoded features. The decoder subsequently reconstructs the video snippet as $R^{N,T,C,H',C'}$, where T is the number of frames per snippet, C color channels, (H', W') reconstructed frame dimensions. With the reconstructed frames, we take first $T-1$ frames to predict the T^{th} frame ($\bar{Y}^{T_{pre}} = h_{Tpre}(f(S_i))$).

3.4. Self-Supervised Learning Task

In our approach, we design a pretext task for each type of augmentation. To solve these self-supervised tasks, we apply multiple task-specific heads, h_{Tn}, which take snippet-level features, $f(S_{Tn})$, as input. Each self-supervised task is treated as a classification problem, and all tasks are considered equally important for incorporating temporal context information. The ground truth labels for the four tasks are as follows:

- **Temporal Reverse**: A binary classification indicating whether the snippet is reversed or not.
- **Temporal Speed**: A binary classification distinguishing between accelerated frames and consecutive frames.
- **Temporal Shuffle**: A binary classification indicating whether the frames in the snippet are shuffled or not.
- **Jittered Rotation**: A four-class classification for four types of rotations (0°, 90°, 180°, 270°).

The loss computation is performed jointly by combining the losses from all four task heads, and we utilize cross-entropy loss for learning each self-supervised task.

$$\mathcal{L}_{T_n}(S^{T_n}, Y^{T_n}) = -\sum_{k=1}^{K} Y_k^{T_n} \log\left(\hat{Y}_k^{T_n}\right) \tag{2}$$

where $\hat{Y}^{T_n} = \text{softmax}\left(h_{T_n}(f(S^{T_n}))\right)$ is the output of the task head h_{T_n}, K is the number of classes, and Y^{T_n} is the one-hot encoding of the ground-truth label for S^{T_n}. For different self-supervised tasks, they are generally considered as equally important for incorporating temporal context information.

$$\mathcal{L}_{\text{st}} = \sum_{n=1}^{N} \mathcal{L}_{T_n}(S^{T_n}, Y^{T_n}) = -\sum_{n=1}^{N}\sum_{k=1}^{K} Y_k^{T_n} \log\left(\hat{Y}_k^{T_n}\right). \tag{3}$$

The Task head network architecture consists of a fully connected layer, a reshaping operation, and a linear layer.This reshapes and maps the encoded features into a format suitable for downstream tasks (e.g., classification or detection). For optimization, we use the Adam optimizer.

3.5. Contrastive Learning Task

We employed Momentum Contrast v2 (MoCo v2) to learn feature representations without labeled data. MoCo v2 uses momentum encoding and contrastive loss to generate embeddings where similar inputs are closer, and dissimilar ones are farther apart in the embedding space. Two augmented views of the same image are used as inputs: the query input (x_q) and key input (x_k). These are 4D tensors of shape (n, 512, 32, 32) where n is the snippet size. Augmentations like cropping, flipping, and color jittering are applied to ensure invariance to transformations.

MoCo v2 uses two encoders: the query encoder processes x_q to produce the query embedding q, and the momentum encoder processes x_k to produce the key embedding k. All encoders share a projection head comprising a fully connected layer, L2 normalization, and a linear layer. This head transforms features into a 128-dimensional embedding space, enhancing representation learning. The momentum encoder's weights are updated using a moving average of the query encoder's weights with the rule:

$$\theta_{\text{momentum}} = \text{momentum}\cdot\theta_{\text{momentum}} + (1-\text{momentum})\cdot\theta_{\text{query}}$$

where the momentum coefficient is set to 0.999 for stability.

MoCo v2 optimizes embeddings using the InfoNCE loss, which aligns embeddings of positive pairs (same image) and separates negative pairs (different images). The query embedding q is compared to all key embeddings k in a mini-batch using cosine similarity. A temperature parameter (0.07) sharpens similarity scores. The loss is defined as:

$$\text{Loss} = -\log\left(\frac{\exp(\text{sim}(q,k)/\tau)}{\sum \exp(\text{sim}(q,k')/\tau)}\right)$$

where sim(q, k) represents cosine similarity, and 'k'' are negative samples.

Key improvements in MoCo v2 include a projection head for better feature learning and stronger augmentations like color jittering and cropping. These enhancements enable MoCo v2 to effectively learn robust representations suitable for downstream tasks.

3.6. Inference

During inference, for each test sample, we calculate the final regularity score of it as the average score given by multiple task heads. For frame prediction, we measure the prediction error between the predicted frame $\bar{Y}$ and the ground truth Y with peak signal to noise ratio(PSNR).

$$S_{Tpre} = 10\log_{10}\frac{max^2(\bar{Y})}{\|Y-\bar{Y}\|_2^2/M} \tag{4}$$

where $max(\bar{(Y)})$ is the maximum of $\bar{Y}$, and M is the pixels number of Y. High values of PSNR indicate that Y is more likely to be normal. For the self-supervised tasks, we regard the average probability of that video snippets are augmented with the corresponding transformations as anomaly scores. The final regularity score of the self-supervised task is calculated as, $S_{Tst} = 1 - \frac{1}{n}\sum_{i=1}^{n} \bar{Y}^{T_n}$. For the instances contrast task, we take the total contrast similarities between features of different augmented snippets as the regularity score S_{Tcon}. The final regularity score is calculated as, $S = S_{Tpre} + S_{Tst} + S_{Tcon}$. Finally, the regularity score is smoothed by a temporal Gaussian filter with $\sigma = 1$.

4. Experiements

4.1. Datasets

UCSD Ped1 dataset was used in which it contains videos of pedestrians walking in a walkway area and Anomalies include non-pedestrian entities like bicycles, vehicles, or people performing unusual actions. This dataset contains 14,000 frames, which are divided into 34(6800 frames) and 36 videos(7200 frames) for training and testing, respectively. There are 12 types of abnormal events, including: carts, cars, skateboarding, and cycling among pedestrians.Videos are grayscale with a resolution of 158 × 238 pixels per frame. Training set contains only frames of normal behaviors whereas test set contains both normal and anomalous frames.

4.2. Data Augmentation

We formed the snippets S_i by stacking 8 consecutive frames with non-overlapping between snippets. Each snippet is transformed with the four temporal transformations mentioned in the methodology. We can see the jittered rotation of a frame as shown in the figure 4. A video folder consists of 200 frames, so we get a total of 200/8 = 25 snippets. And Each frame is with a resolution of $3 \times 158 \times 238$. So the dimensions of input is given by $\mathbb{R}^{25\times8\times3\times158\times238}$.

4.3. Implementation Details

All methods were implemented using the Python programming language on a system with 32GB of RAM and an NVIDIA T400 GPU, which features 2GB of dedicated memory.

4.4. Supervised Methods

As we aim to familiarize ourselves with the snippet dimensions, we were experimented with well-known supervised methods, considering the test video folders with ground truth. We used basic Support Vector Machine (SVM) [1], Random Forest Classifier (RFC) [23], and XgBoost [22] architectures for evaluation.The performance was tabulated in Table.1

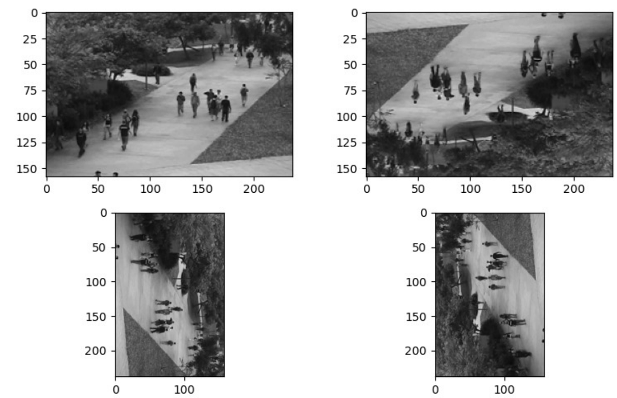

Figure 4. Jittered Rotation Augmentation

Method	Accuracy (%)
SVM	79.63
RFC	77.89
XgBoost	68.89

Table 1. Comparison of Methods Based on Accuracy

4.5. Training

The training process involved preprocessing input videos to a resolution of 256 × 256, with pixel intensity values normalized to the range [-1,1]. For consistency, the videos were divided into snippets, with each snippet consisting of 8 consecutive frames. Because of device configuration limitations,only one training video was considered. The training process was conducted over 50 epochs. The parameter t is set to 8 (Number of frames per snippet).

5. Discussion

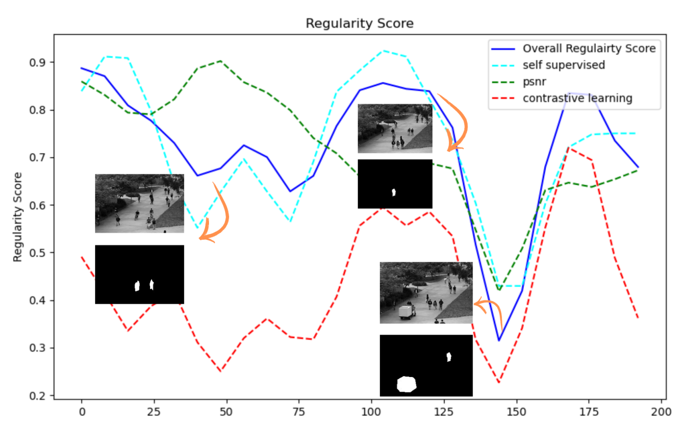

Figure 5. Overall regularity score

5.1. Ablation Study

We conduct experiments to investigate the contribution of each component. For frame prediction, the figure 6 is the PSNR values of frames during frame prediction. The green line represents the PSNR values for video reconstructions and the blue line represents the PSNR values for frame prediction. As we can see that, frame prediction is able to distinguish abnormal frames rather than the video reconstruction.

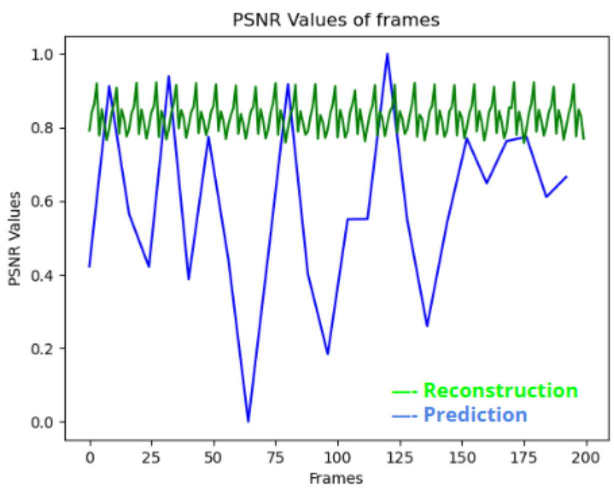

Figure 6. Comparison of PSNR values between Frame Reconstruction and Prediction.

For regularity score to determine the abnormality, we consider $S = S_{Tpre} + S_{Tst} + S_{Tcon}$. we evaluated the contribution of each individual component and their combinations toward the final regularity score. For each combination, we computed the regularity scores and generated the corresponding regularity plots. The figure 9 were analyzed to evaluate their ability to distinguish between normal and abnormal events effectively. $(S_{Tpre}, S_{Tcon}, S_{Tst}, S_{Tpre} + S_{Tcon}, S_{Tpre} + S_{Tcon}, S_{Tst} + S_{Tcon})$.

5.2. Visualization and Analysis

The Figures 5, 7, 8 shows the overall Regularity scores of the test video, from which the solid blue line represents the total regularity score where as the dotted lines represents the regularity scores obtained from frame prediction, self-supervised and contrastive learning tasks. As we can see the regularity scores drop down when abnormal event happens. Specifically the regularity score curves drop sharply when abnormal objects suddenly appear and continuously remain at a quite low level when the abnormal events are in progress. When the abnormal objects disappear or the abnormal behaviors are over, the curves quickly rise to a fairly high level.

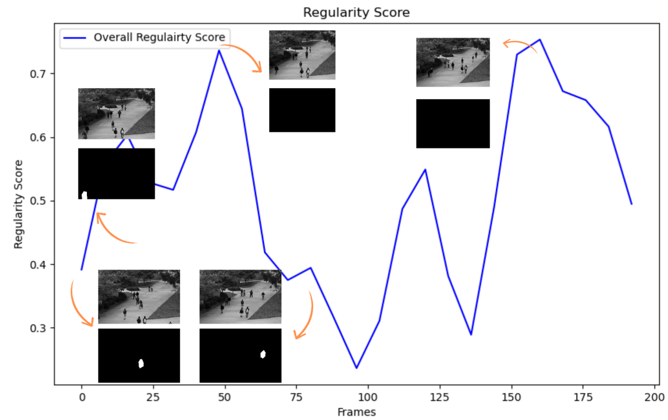

Figure 7. Overall regularity scores of 2nd test sample

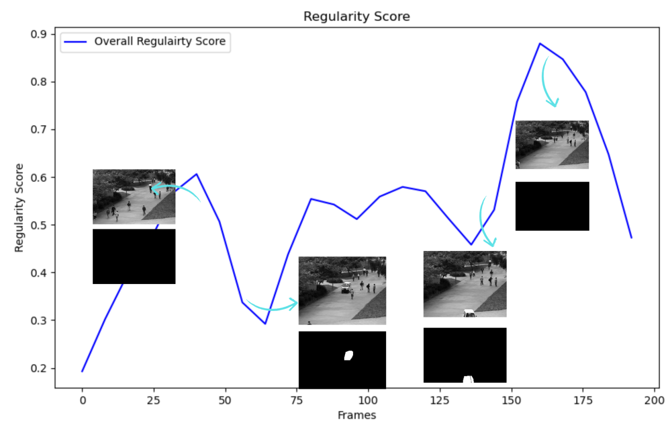

Figure 8. Overall regularity scores of 3rd test sample

6. Conclusion

In this work, we proposed an architecture for VAD that integrates frame detection, self-supervised learning, and contrastive learning. The use of a multi-task learning approach significantly enhances the model's ability to accurately predict anomalies. Spatial-temporal transformations were employed to create multiple tasks for self-supervised and contrastive learning. While our model was trained using non-overlapping video snippets, future work could benefit from using overlapping snippets to better capture boundary anomalies and reduce missed events. Additionally, expanding the dataset with videos from diverse domains is crucial for achieving improved generalization and model performance.

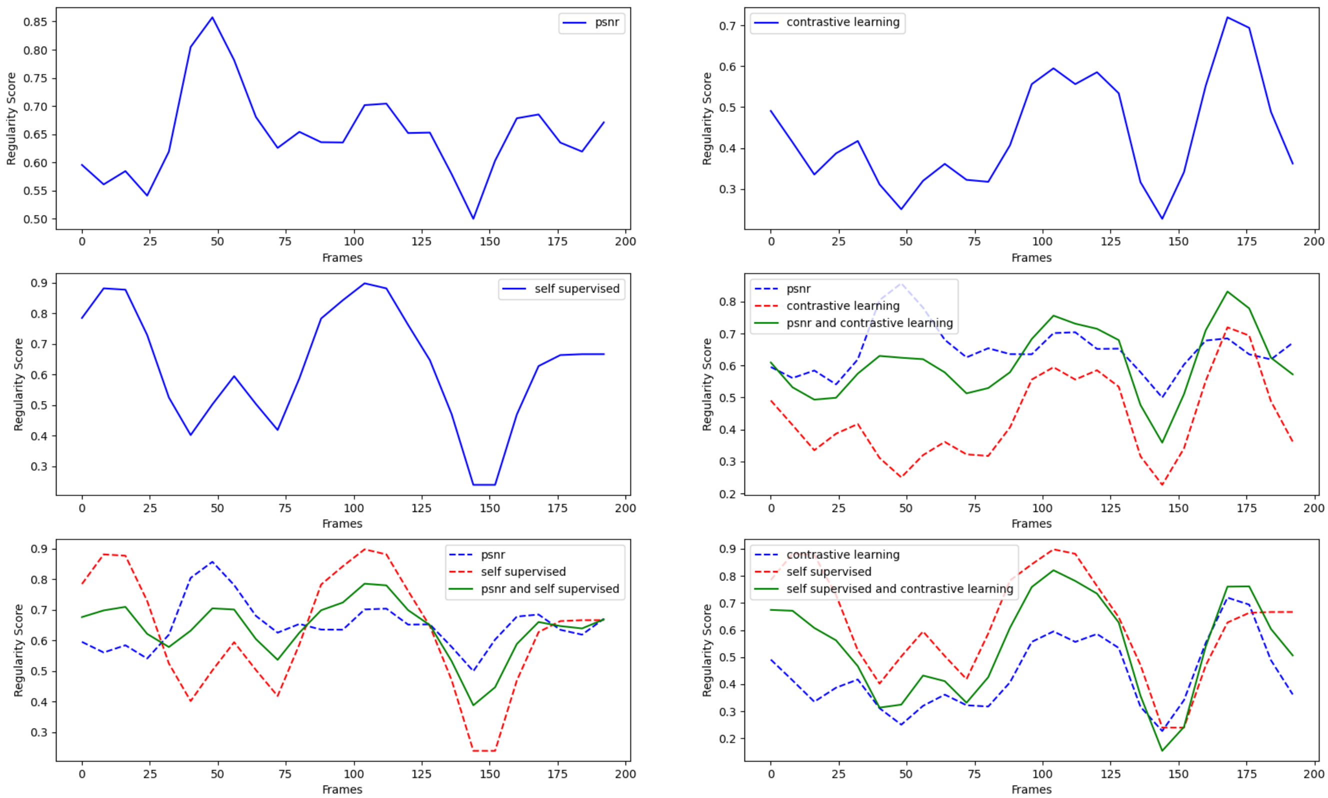

Figure 9. Various combinations of scores

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